A Lightweight Transformer Network for Hyperspectral Image Classification

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***Abstract*— Transformer is a powerful tool for capturing long-range dependencies and has shown impressive performance in hyperspectral image (HSI) classification. However, such power comes with a heavy memory footprint and huge computation burden. In this article, we propose two types of lightweight self-attention modules (a channel lightweight multihead self-attention (CLMSA) module and a position lightweight multihead self-attention (PLMSA) module) to reduce both memory and computation while associating each pixel or channel with global information. Moreover, we discover that transformers are ineffective in explicitly extracting local and multiscale features due to the fixed input size and tend to overfit when dealing with a small number of training samples. Therefore, a lightweight transformer (LiT) network, built with the proposed lightweight self-attention modules, is presented. LiT adopts convolutional blocks to explicitly extract local information in early layers and employs transformers to capture long-range dependencies in deep layers. Furthermore, we design a controlled multiclass stratified (CMS) sampling strategy to generate appropriately sized input data, ensure balanced sampling, and reduce the overlap of feature extraction regions between training and test samples. With appropriate training data, convolutional tokenization, and LiTs, LiT mitigates overfitting and enjoys both high computational efficiency and good performance. Experimental results on several HSI datasets verify the effectiveness of our design.**

***Index Terms*— Deep learning (DL), hyperspectral image (HSI) classification, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain tens or even hundreds of narrow and continuous spectral bands ranging from visible to infrared [1]. The abundant spectral information in HSIs enables object identification and detection at fine-grained spectral scales, making HSIs useful in a wide range of applications, such as forest monitoring [2], medical imaging [3], and urban development observation [4]. One of the fundamental tasks in these applications is classification, which involves assigning a specific category label to each pixel.

Traditional classifiers focus only on spectral information, including support vector machine (SVM) [5], random forest [6], and logistic regression [7]. Complex scenarios and spectral heterogeneity of objects require to model spatial correlations. This can be achieved by different approaches, such as Gabor filtering [8], morphological profiles [9], gray-level co-occurrence matrix [10], and 3-D wavelets [11]. As an alternative, kernel-based methods [12], [13] have also been introduced to capture spatial information of HSIs. However, these traditional approaches rely heavily on prior knowledge and are limited to shallow features, resulting in poor generalization and robustness [14].

Deep learning (DL), an automatic feature learning technique, can automatically learn richer representations than traditional approaches. Several DL techniques have been applied to HSI classification, such as deep belief networks [15], convolutional neural networks (CNNs) [16], graph convolutional networks (GCNs) [17], [18], and transformers [19]. Among these DL techniques, CNNs are the most widely used due to their inherent properties of local connectivity and weight sharing. These properties impose strong constraints on convolution weights that hardcode inductive biases into networks, thus leading to more sample-efficient and parameter-efficient [20].

Many works [14], [21], [22], [23] have attempted to bring the power of CNNs to HSI classification. The existing methods mainly fall into two broad categories: patch-based classification and fully convolutional network (FCN)-based segmentation frameworks. Spectral–spatial DL networks, such as spectral–spatial residual network (SSRN) [21], dual-level deep spatial manifold representation network (SMRN) [24], and spectral–spatial self-attention network (SSSAN) [14], follow the design rule of patch-based classification to facilitate feature learning and classifier training. Features are extracted from the spatial patch centered on the sample pixel and further processed to assign a specific category to the center pixel. However, redundant computation on overlapping regions between adjacent patches is inevitable, which severely hampers large-scale applications. In contrast, FCNs exhibit better accuracy–speed trade-offs. FCNs feed data into the network and perform feature extraction and pixel-to-pixel classification. Examples of these models are the

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spectral–spatial FCN (SSFCN) [22] and encoder–decoder architectures [23]. Although these CNN-based architectures can learn information from a larger receptive field by stacking multiple layers, they still lack global connectivity. Moreover, convolutional filter weights are usually fixed after training and cannot be dynamically adapted to different inputs [25].

Recently, transformers have shown promising results in various visual tasks, such as image classification [26] and semantic segmentation [27]. As a core component of transformers, self-attention has the properties of long-range modeling and adaptive spatial aggregation. Benefiting from the flexible self-attention, transformers can learn more robust and accurate representations than CNNs from extensive data [28]. Vision transformer (ViT) [29] is the first pure transformer backbone for vision tasks. It replaces the inductive biases inherent in convolution with global processing driven by self-attention, demonstrating that large-scale training can trump inductive biases. Several studies have investigated the use of transformers for HSI classification. SpectralFormer [19] generates groupwise spectral embeddings by learning local sequence information from adjacent bands of HSIs. A spectral–spatial transformer network [30] employs consecutive spatial and spectral transformer blocks to learn spectral–spatial information from input patches. To fully exploit the abundant spectral information in HSIs, a two-branch pure transformer [31] uses a spectral transformer and a spatial transformer to learn spectral sequences and spatial features, respectively. A multilevel spectral–spatial transformer network [32] processes HSIs into sequences, learns feature representations with a pure transformer encoder, and then processes multilevel features with decoders to produce classification maps. Huang et al. [33] improved the original Swin transformer [34] to the 3-D structure and proposed a 3-D Swin transformer to explore the rich spatial–spectral information of HSIs. However, these transformer architectures are still inferior to their CNN counterparts for training on small-scale datasets, as they lack proper inductive biases and, thus, require substantial data and computational resources to compensate [25]. Consequently, CNNs are still the preferred models for HSI classification, since they require less time, data, and memory for training, whereas they do not enjoy long-range modeling [35].

Hybrid architectures combining transformers and convolutions have received much attention in constructing lightweight, high-performance models [36]. Hyper-ES^{2}T [37] embeds convolution layers before each spatial–spectral transformer block. In [38], 3-D convolution is embedded in a two-branch transformer to capture global–local dependencies in both spectral and spatial domains. Grouped multi-attention network (GMA-Net) [39] is also a two-branch architecture, where one branch is responsible for spectral–spatial feature learning using CNN and multiattention modules, and the other for pixelwise spectral feature learning using convolutions. Some works [35], [40], [41] first employ convolution to perform shallow feature extraction and use transformers to capture the global relationship between different tokens. Hyperspectral image transformer (HiT) [42] embeds convolution operations into the transformer structure. It uses 3-D convolution layers to produce local spatial–spectral information and then uses depthwise and pointwise convolution operations to encode spatial–spectral representations along height, width, and spectral dimensions, respectively.

Although these transformer-based networks have shown impressive performance in HSI classification, the high dimensionality, together with limited labeled samples, still makes the classification task very challenging. There are still three main aspects that need to be addressed.

1) The existing transformer-based models mainly follow the patch-based classification, which not only results in redundant computation but also hinders long-range spatial dependency modeling. Thus, an excellent approach to build effective architectures is to follow the FCN-based framework.
2) Self-attention in transformers incurs a heavy memory footprint and huge computation burden, which limits their real-time applications.
3) Transformers are ineffective in explicitly extracting local and multiscale features and are prone to overfitting with small numbers of training samples.

To overcome the above three drawbacks, we propose a lightweight transformer (LiT) network and a controlled multiclass stratified (CMS) sampling strategy. Specifically, LiT is proposed to merge and exploit the advantages of transformers (i.e., long-range dependence and input adaptive weighting) and CNNs (i.e., local spatial modeling and inductive bias) for end-to-end pixel-to-pixel classification by following the FCN-based framework. LiT has four stages to generate feature maps with different scales. The first two stages deploy convolutional blocks to explicitly extract local information, and the last two stages employ transformer blocks to capture long-range dependencies. In LiT, two lightweight self-attention modules are designed to reduce both computation and memory while associating each pixel/channel with global information. The CMS sampling strategy is designed to provide appropriately sized inputs and ensure balanced sampling while reducing the overlap of feature extraction regions between training and test samples. The proposed LiT combined with the CMS sampler can mitigate overfitting. Superior performance on three HSI datasets demonstrates the effectiveness of our approaches.

The main contributions of this study are summarized as follows.

1) Extending transformer-based research to FCN-based HSI classification by introducing LiT, which can incorporate the merits of transformers and CNNs to efficiently model image hierarchies in local and long range, achieving high performance and fast inference speed.
2) Developing two simple yet effective self-attention modules—the position lightweight multihead self-attention (PLMSA) module and the channel lightweight multihead self-attention (CLMSA) module—to construct compact long-range dependencies in spatial and spectral dimensions, respectively, resulting in performance boosts with less computational and memory cost.
3) Designing a CMS sampling strategy to provide appropriately sized input data and ensure balanced sampling. It can also significantly reduce the overlap of feature extraction regions between training and test samples, providing a more objective benchmark for evaluating method performance.

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The remainder of this study is organized as follows. Section II reviews transformer-related work and classical CNN-based networks for HSI classification. Section III describes the proposed LiT network and CMS sampling strategy in detail. Extensive experiments with ablation studies are performed and discussed in Section IV. Finally, some concluding remarks and a brief outlook on future research are presented in Section V.

II. Related Works

A. Vision Transformer

ViT [29] provides an alternative design paradigm to CNNs. It reshapes the input image into a sequence of flattened patches, which are then projected into a sequence of patch embeddings using a trainable linear projection. These patch embeddings append a class token and then incorporate position embeddings to obtain a sequence of input tokens $\mathbf{z}_0$.

The transformer encoder, consisting of L transformer layers, is applied to $\mathbf{z}_0$ to learn interpatch representations. A transformer layer consists of a multihead self-attention (MSA) block (1) followed by a feed-forward network (FFN) (2). Layer normalization (LN) is applied before each block, and a shortcut connection is added after each block. Given input $\mathbf{z}_{l-1}$, a transformer layer can be written as follows:

$$\mathbf{z}'_l = \mathrm{MSA}(\mathrm{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1} \tag{1}$$

$$\mathbf{z}_l = \mathrm{FFN}\left(\mathrm{LN}\left(\mathbf{z}'_l\right)\right) + \mathbf{z}'_l \tag{2}$$

where $l = 1, \ldots, L$. The MSA module is defined by considering h "heads." Specifically, the input $\mathbf{x}_{\mathrm{in}}$ is divided equally into h "heads" by channel (3), and the self-attention function is applied to each "head" (4). These h sequences are then concatenated (5)

$$\mathbf{x}_{\mathrm{in}} \rightarrow \left\{x^1_{\mathrm{in}}, \ldots, x^h_{\mathrm{in}}\right\} \tag{3}$$

$$h(\mathbf{x}_i) = \mathrm{Softmax}\left(\mathbf{Q}_i \mathbf{K}_i^T \Big/ \sqrt{d_i^k}\right)\mathbf{V}_i \tag{4}$$

$$\mathrm{MSA}(\mathbf{x}_{\mathrm{in}}) = \mathrm{Concat}\left[h\left(x^1_{\mathrm{in}}\right)_1, \ldots, h\left(x^h_{\mathrm{in}}\right)_h\right] \tag{5}$$

where $\mathbf{Q}_i = \mathbf{x}^i_{\mathrm{in}} W^i_Q$, $\mathbf{K}_i = \mathbf{x}^i_{\mathrm{in}} W^i_K$, and $\mathbf{V}_i = \mathbf{x}^i_{\mathrm{in}} W^i_V$; W^i_Q, W^i_K, and W^i_V are the learnable projection matrices to project the $\mathbf{x}^i_{\mathrm{in}}$ into different feature spaces, $i = 1, \ldots, h$. The FFN consists of two linear layers separated by an activation function, i.e.,

$$\mathbf{Y} = f(\mathbf{Z}\mathbf{W}_1)\mathbf{W}_2 \tag{6}$$

where $\mathbf{W}_1 \in \mathbb{R}^{c \times \gamma c}$, $\mathbf{W}_2 \in \mathbb{R}^{\gamma c \times c}$, $f(\cdot)$ denotes an activation function, and $\mathbf{Y}$ is the output of FFN. c is the channel dimension of $\mathbf{Z}$, and γ is the dimension expansion ratio, usually set to 4. The bias term is omitted for simplicity.

ViT is computationally intensive, challenging to optimize, and requires extensive data to avoid overfitting due to the absence of inductive bias [25]. Various design strategies have been explored to incorporate the advantages of CNNs into transformer models to improve performance. To generate hierarchical representation and reduce computational cost, hierarchical visual transformer (HVT) [43] gradually pools visual tokens to reduce the sequence length as the layer goes deeper, which is similar to downsampling in CNNs. CoAtNet [25] improves the generalization and capacity of the model by naturally unifying and summing depthwise convolution and self-attention through simple relative attention. Some works [25], [44] apply convolution before transformer layers to generate richer tokens and preserve local information. Mobile-former [45] is a bridge-connected parallel architecture of MobileNet and transformer, combining the advantages of MobileNet for local processing and transformer for global interaction. CNNs meet transformers (CMT) [26] embeds the depthwise convolution into transformer blocks to enhance local information. Li et al. [46] introduce locality guidance provided by a trained CNN to accelerate convergence and improve the performance of ViTs on tiny datasets. MobileViT [36] is a general and lightweight ViT for mobile devices, where the MobileViT block replaces local convolutional processing with global processing using transformers, allowing it to have ViT- and CNN-like properties.

Furthermore, many efficient attention mechanisms [47], [48], [49] have been developed to boost performance and save computational cost. One way is to restrict self-attention to small windows, such as Swin transformer [34]. Besides, BiFormer [47] designs a novel bi-level routing attention to save computation and memory, where each query focuses on a small subset of the most semantically relevant key–value pairs. EfficientViT [48] presents a cascaded group attention module that feeds attention heads with different splits of the full features to reduce computational redundancy. A super token attention mechanism [49] is designed to promote effective and efficient global context modeling at the early stages of a network.

However, these transformer-based networks are still heavy weight for the HSI classification task. Combining the strengths of CNNs and transformers to build ViT models for HSI classification tasks remains an open question.

B. CNN for HSI Classification

CNNs have become the dominant technique in HSI classification in recent years because of their strong representation, lightweight, and easy optimization. In [16], patches centered on sample pixels are generated and fed into a sequence of convolution and pooling layers, followed by linear layers for feature extraction and classification. A CNN with pixel-pair features was proposed in [50] to improve feature learning. Some networks use parallel filters with different kernel sizes [51] or short connections [52] to promote multiscale information learning. Two-branch CNN-based architectures [14], [53] employ a 2-D CNN and other algorithms (e.g., 1-D CNN and stacked autoencoders) to learn spatial and spectral information, respectively. The 3-D CNNs have been introduced to extract spectral–spatial features for HSI classification considering the 3-D structural characteristics of HSIs [21]. These patch-based classifications have difficulties to achieve fast inference speed due to redundant computation of overlapping regions between adjacent patches, which limits practical applications. FCN-based segmentation networks [22], [23], [54], [55] mitigate the redundant computation by performing pixel-to-pixel classification. For example, SSFCN [22] and a deep FCN with an efficient nonlocal module (ENL-FCN) [54] take an entire HSI as input and perform feature extraction without reducing spatial dimensions. FreeNet [23] and spectral–spatial-dependent global learning (SSDGL) [55] networks follow

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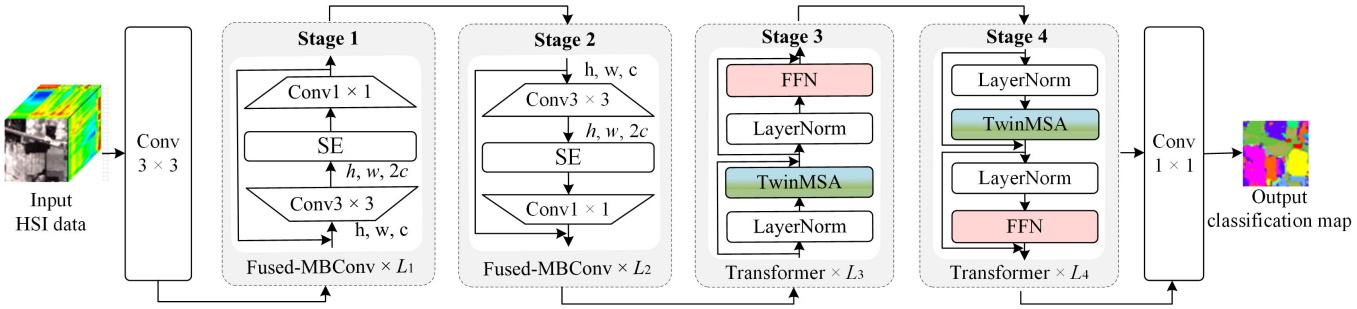


Fig. 1. Overview illustration of the proposed LiT network for HSI classification.

the encoder–decoder architecture to perform pixel-to-pixel classification.

III. Methodology

This section introduces the proposed approaches: the LiT network and the CMS sampling strategy.

A. LiT Network

This study develops a simple yet effective network, LiT, which aims to improve the feature representation for HSI classification. In ViT, linear projections poorly model the structural information present in patches. To alleviate this problem, we use convolutional tokenization to generate richer tokens and preserve more local information. Our LiT has four stages to produce the hierarchical representation of data, as illustrated in Fig. 1. The first two stages deploy fused-MBConv blocks to introduce inductive bias and extract local features. The last two stages employ a sequence of transformer blocks to model long-range dependencies. The CNN stages are implemented before the transformer stages based on the prior knowledge that convolution is good at encoding local information, which is essential for processing low-level features [56]. Considering the redundancy of spectral information in HSIs, a dimensionality reduction convolution is applied at the end of each stage to aggregate local spectral features. This can preserve more details without defining additional parameters. In this way, LiT improves context aggregation for better identification and refines coarse results. The output of stage 4 is fed into a 1×1 convolutional layer for classification. In the transformer block, a twin MSA (TwinMSA) module is designed to promote spatial–spectral information learning and reduce the high computational and memory cost of traditional self-attention. As shown in Fig. 2, the TwinMSA consists of PLMSA and CLMSA modules. Details of the Fused-MBConv block, PLMSA, and CLMSA modules are presented below.

1) Fused-MBConv Block: Many state-of-the-art (SOTA) LiT models, such as CoAtNet [25] and MobileViT [36], use the inverted residual block for efficiency. The inverted residual block first widens channels with a 1×1 expansion convolution, and then uses a 3×3 depthwise convolution to capture local information. Furthermore, it uses a 1×1 convolution to project the channel dimension to the original size, so that input and output can be added. Although depthwise convolutions have fewer parameters and floating-point operations (FLOPs) than regular convolutions, they are slow in the early stages, because they cannot fully utilize modern accelerators [57]. Fused-MBConv replaces the 1×1 expansion convolution and 3×3 depthwise convolutions with a single 3×3 expansion convolution. As illustrated in Fig. 1, Fused-MBConv consists of a 3×3 expansion layer, a squeeze-and-excitation (SE) module [58], and a 1×1 projection layer. The expansion ratio of the 3×3 expansion convolution is set to 2, and the 1×1 convolution reduces the channel dimension by the same ratio. The input and output of Fused-MBConv block are connected when they have the same number of channels. Compared with the original ViT, the convolutional tokenizer that adopts Fused-MBConv in the early stages is more effective at encoding spatial information.

2) Position Lightweight Multiheaded Self-Attention: The computational complexity of self-attention in transformers is quadratic to the spatial size of inputs. Using self-attention modules to process high-resolution images would inevitably cause the problem of low computational efficiency and insufficient memory. To alleviate this problem, we propose the PLMSA module that constructs the relationships between each feature vector and eigenvector clusters. The eigenvector cluster is a compact feature vector collected from a subset of feature vectors in input tensors. They are implemented using the spatial pyramid pooling-fast technique [59], which reduces computational burden and provides multiscale contextual information.

As illustrated in Fig. 2, the input token $\mathbf{X} \in \mathbb{R}^{w \times h \times d}$ is reshaped into $\hat{\mathbf{X}} \in \mathbb{R}^{n \times d}$, where w, h, and d represent the height, width, and channels of $\mathbf{X}$, respectively, $n = wh$. Meanwhile, we use 2×2 max pooling with stride 2 to reduce the spatial size of $\mathbf{K}^p$ and $\mathbf{V}^p$ before the self-attention operation. Specifically, we feed $\mathbf{X} \in \mathbb{R}^{w \times h \times d}$ into two successive max-pooling layers to generate $\mathbf{X}_1^p \in \mathbb{R}^{(w/2) \times (h/2) \times d}$ and $\mathbf{X}_2^p \in \mathbb{R}^{(w/4) \times (h/4) \times d}$, respectively. $\mathbf{X}_1^p$ and $\mathbf{X}_2^p$ are separately reshaped into $\mathbb{R}^{(n/4) \times d}$ and $\mathbb{R}^{(n/16) \times d}$, which are concatenated to form $\mathbf{X}^p \in \mathbb{R}^{(5n)/16 \times d}$. In addition, we inject positional information into each attention block by introducing a relative bias $\mathbf{B}^p$ to the attention maps, and the corresponding position lightweight self-attention (PLSA) is defined as follows:

$$\text{PLSA} = \text{Softmax}\left(\mathbf{Q}^p \left(\mathbf{K}^p\right)^{\mathrm{T}} \Big/ \sqrt{d_k} + \mathbf{B}^p\right)\mathbf{V}^p \tag{7}$$

where $\mathbf{Q}^p = \hat{\mathbf{X}} W_Q \in \mathbb{R}^{n \times d}$, $\mathbf{K}^p = \mathbf{X}^p W_K^p \in \mathbb{R}^{(5n)/16 \times d}$, $\mathbf{V}^p = \mathbf{X}^p W_V^p \in \mathbb{R}^{(5n)/16 \times d}$, and $\mathbf{B}^p \in \mathbb{R}^{n \times (5n)/16}$. Finally, the PLMSA

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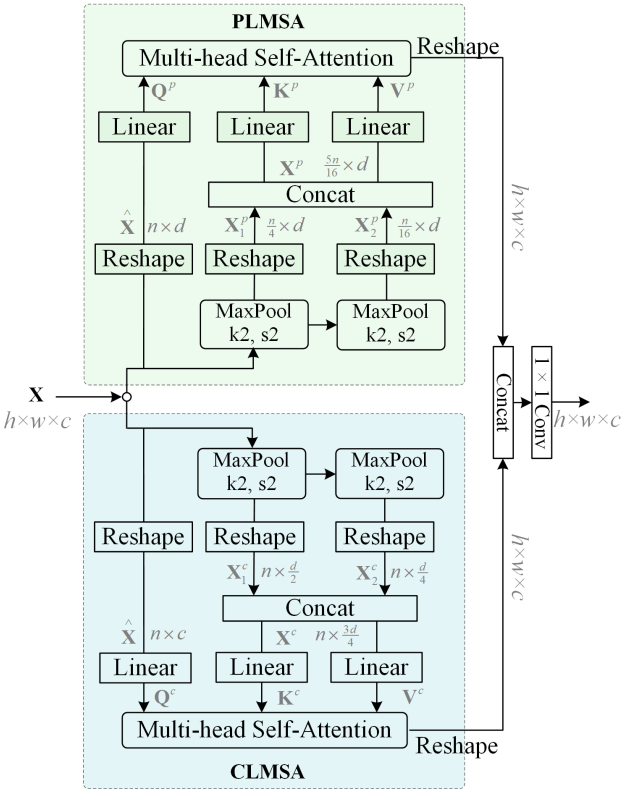


Fig. 2. Illustration of the TwinMSA module, which consists of PLMSA and CLMSA.

module is defined by considering h "heads," i.e., the inputs $\hat{\mathbf{X}}$ and $\mathbf{X}^p$ are divided equally into h "heads" by channel, and the PLSA function is applied to each "head." Each "head" outputs a sequence of size $n \times (d/h)$. These h sequences are then concatenated into a sequence of $n \times d$. Details of the operation of MSA can be found in (3)–(5).

3) Channel Lightweight Multiheaded Self-Attention: Each deep-level feature channel can be viewed as a class-specific response, and these semantic responses are correlated [60]. It is possible to enhance feature representation of related semantics by exploiting dependencies between different channel maps. In addition, given the abundant spectral information in HSI, a high degree of redundancy between adjacent channels is inevitable. To better exploit long-range spectral information with less computation, we propose the CLMSA module. CLMSA builds the relationship between each channel and each channel cluster, which is a compact channel map. Channel clusters are generated using the pyramid pooling-fast technique along the channel dimension to aggregate channel maps from input tensors.

As shown in Fig. 2, we first reshape $\mathbf{X} \in \mathbb{R}^{w\times h\times d}$ to $\hat{\mathbf{X}} \in \mathbb{R}^{n\times d}$ while using max pooling with stride 2 along the channel axis to reduce the spectral dimension of $\mathbf{K}^c$ and $\mathbf{V}^c$ before the self-attention operation. Specifically, we feed $\mathbf{X} \in \mathbb{R}^{w\times h\times d}$ into two successive max-pooling operations to generate $\mathbf{X}_1^c \in \mathbb{R}^{w\times h\times (d/2)}$ and $\mathbf{X}_2^c \in \mathbb{R}^{w\times h\times (d/4)}$, which are then reshaped into $\mathbf{X}_1^c \in \mathbb{R}^{n\times (d/2)}$ and $\mathbf{X}_2^c \in \mathbb{R}^{n\times (d/4)}$, respectively. $\mathbf{X}_1^c$ and $\mathbf{X}_2^c$ are then concatenated to form $\mathbf{X}^c \in \mathbb{R}^{n\times (3/4)d}$. Each channel map of $\mathbf{X}_1^c$ and $\mathbf{X}_2^c$ is viewed as a channel cluster. A relative bias $\mathbf{B}^c$ is added to the attention map to provide positional information. The channel lightweight self-attention (CLSA) is calculated as follows:

$$\mathrm{CLSA} = \mathrm{Softmax}\left(\mathbf{Q}^c\left(\mathbf{K}^c\right)^{\mathrm{T}}\Big/\sqrt{d_k} + \mathbf{B}^c\right)\mathbf{V}^c \tag{8}$$

where $\mathbf{Q}^c = \hat{\mathbf{X}}$, $\mathbf{K}^c = \mathbf{X}^c$, and $\mathbf{V}^c = \mathbf{X}^c$. The input features are not transformed with learnable projection matrices to maintain the relationships between different channel maps before the self-attention map is computed. Similarly, the CLMSA module is also defined by considering h "heads." Specifically, the input is divided equally into h "heads," and the CLSA function is applied to each "head." The output of each "head" is then concatenated.

To take full advantage of long-range contextual information in spectral and spatial dimensions, we concatenate the outputs of PLMSA and CLMSA modules and apply a 1×1 convolution layer to project the channel dimension to the input size $w \times h \times d$, as indicated in Fig. 2. Before the concatenation operation, the outputs of PLMSA and CLMSA are reshaped into $w \times h \times d$

$$\mathrm{TwinMSA} = f([\mathrm{PLMSA}, \mathrm{CLMSA}]) \tag{9}$$

where $[\cdot]$ refers to the concatenation operation and $f(\cdot)$ denotes the 1×1 convolution. PLMSA and CLMSA modules can effectively improve feature representations with minimal parameters and can be directly inserted into the existing backbones. Note that the pooling stride and the number of pooling layers in both modules can be adjusted according to the size of input data. The transformer block, as displayed in Fig. 1, can be formulated as follows:

$$\tilde{\mathbf{z}} = \mathrm{TwinMSA}(\mathrm{LN}(\mathbf{z})) + \mathbf{z} \tag{10}$$

$$\mathbf{y} = \mathrm{FFN}\left(\mathrm{LN}\left(\tilde{\mathbf{z}}\right)\right) + \tilde{\mathbf{z}} \tag{11}$$

where $\mathbf{z}$ and $\mathbf{y}$ are the input and output of the transformer block, respectively. In the FFN, the expansion ratio is reduced from 4 to 2 for better efficiency, and the linear layers are replaced by 1×1 convolution layers.

B. Controlled Multiclass Stratified Sampling Strategy

HSI datasets typically contain only one partially labeled image. In spatial-based classification methods, the commonly used pixel-based random sampling strategy results in overlapping feature extraction spaces between training and test samples. Consequently, information from training data is used to evaluate methods during the testing phase. The improved classification accuracy of spatial-based methods is partly due to the increased dependency between training and test data, compared with pure spectral-based methods [61]. Therefore, this section develops the CMS sampling strategy to alleviate this drawback.

As detailed in Fig. 3, the CMS sampler first partitions the entire HSI into nonoverlapping windows (Step 1). The window size should ensure that each class exists in at least two windows to guarantee that all classes are present in both training and test sets. The CMS sampler will pad the HSI when the height and width cannot be divided by window size. Windows where all pixels have the same labeled class are used as test data, and windows where all pixels are unlabeled are waiting for prediction (Step 2). The next step will

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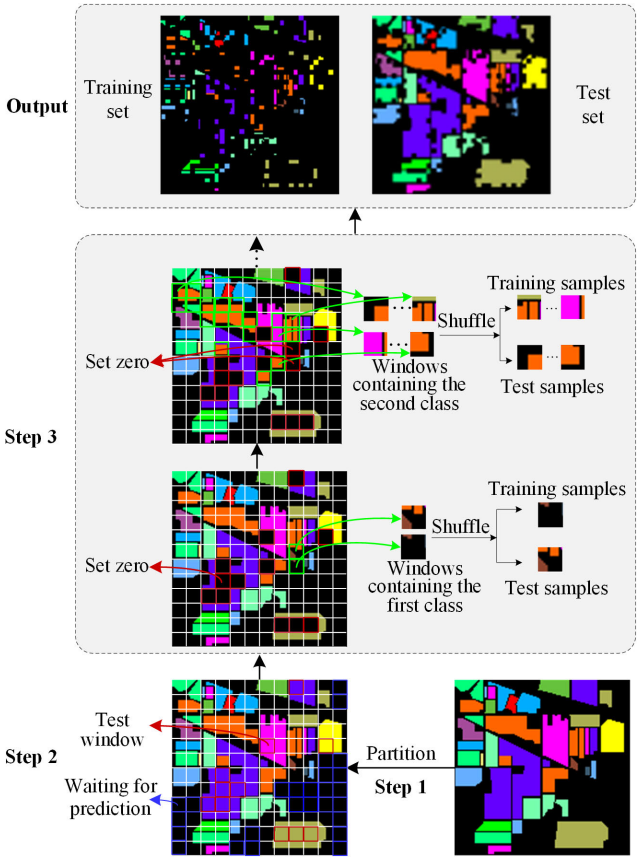


Fig. 3. Illustration of the proposed CMS sampling strategy.

divide the remaining windows with more than one category (including the unclassified ones) into training and test data according to a predefined order. The predefined order can be either by category or by the number of samples within each category.

Take the predefined order by category as an example. As shown in Step 3 of Fig. 3, multiclass windows containing the first class are collected; then, a predefined proportion of windows is randomly selected for training, while the remaining windows are used for testing. Afterward, the labels of all pixels in windows containing the first class are set to zero to avoid repeated sampling. The remaining windows are then used to collect the second category. This process is repeated until sampling is complete for all classes. Finally, all training and test windows form the training and test sets, respectively. Algorithm 1 gives a pseudo-code of our CMS sampling strategy.

The generated windows can be fed directly into FCN-based networks for training or testing. In this case, the feature extraction regions of the training and test samples are independent of each other. Moreover, the input data size can be adjusted to fit the requirements of the actual situation. Compared with random sampling, the sample distribution generated by the CMS sampler is more similar to that of practical applications. Meanwhile, training samples are still distributed throughout the image, covering a wide range of spectral variants. The impact of test data during the training stage can be significantly reduced. The experimental setup with the proposed CMS sampler can help us more objectively evaluate the performance of spatial-based methods.

Algorithm 1 CMS Sampling Strategy

Input: Hyperspectral image (HSI),
HSI size: height (H), width (W),
window size w, sampling rate per class λ
Output: training set and test set

```
if H & W cannot be divided by w then
    Perform the padding operation on the HSI.
end
Split the HSI into non-overlapping w × w windows.
for each window in HSI do
    if all pixels have the same labeled class then
        Classified as test samples.
        Set the labels of all pixels in this window to zero.
    end
    if windows with more than one category then
        Classified as multi-calss windows.
    end
end
for each class c in multi-calss windows do
    Collect windows containing c from multi-calss windows.
    Randomly select λ windows as training samples and the
     rest as test samples.
    Set the labels of all pixels in these window to zero.
end
Training samples and test samples are combined to form the
 training set and test set, respectively.
```

In the CMS sampler, windows where all pixels have the same labeled class can also be used as training samples, called the controlled random stratified (CRS) sampling strategy. However, the experimental results in Section IV-D show that the CMS sampler works better.

IV. Experiments and Results

A. Description of Datasets

We conducted experiments on three different-sized benchmark datasets to evaluate the performance and generalizability of our approaches: Indian Pines, Salinas, and data fusion contest 2018 (DFC 2018).

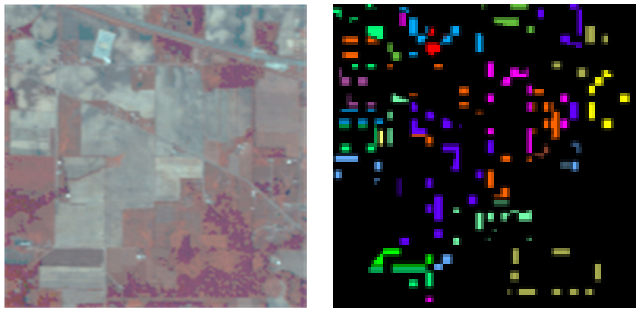
1) Small Dataset: The Indian Pines dataset, covering northwestern Indiana, USA, was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1992. This dataset consists of a 145 × 145 × 224 data cube with a wavelength range of 0.4–2.5 μm and a spatial resolution of 20 m. After discarding 24 noisy and water-absorbing bands, 200 bands were used. There are 16 land-cover classes in the ground truth (GT). Fig. 4(a) summarizes the type and number of samples, and Fig. 4(b) and (c) shows the spatial distribution of training and test data, respectively.

2) Medium Dataset: The Salinas dataset was recorded over several agricultural fields in Salinas Valley, CA, USA, by the AVIRIS sensor. The related map contains 512 × 217 pixels with a spatial resolution of 3.7 m per pixel and 224 bands covering a spectral range of 0.36–2.5 μm. Before experiments, 20 noise and water absorption bands were excluded. Sixteen land-cover classes were defined in the corresponding GT. The type and number of samples are given in Fig. 5(a), and the spatial distribution of training and test data is shown in Fig. 5(b) and (c), respectively.

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ID	Color	Land Cover Type	Train	Test	Total
C1		Alfalfa	14	32	46
C2		Corn-notill	249	1179	1428
C3		Corn-mintill	195	635	830
C4		Corn	58	179	237
C5		Grass-pasture	107	376	483
C6		Grass-trees	146	584	730
C7		Grass-pasture-mowed	12	16	28
C8		Hay-windrowed	94	384	478
C9		Oats	8	12	20
C10		Soybean-notill	164	808	972
C11		Soybean-mintill	380	2075	2455
C12		Soybean-clean	151	442	593
C13		Wheat	73	132	205
C14		Woods	213	1052	1265
C15		Buildings	111	275	386
C16		Stone-Steel-Towers	39	54	93
		Total	2014	8235	10249

(a)

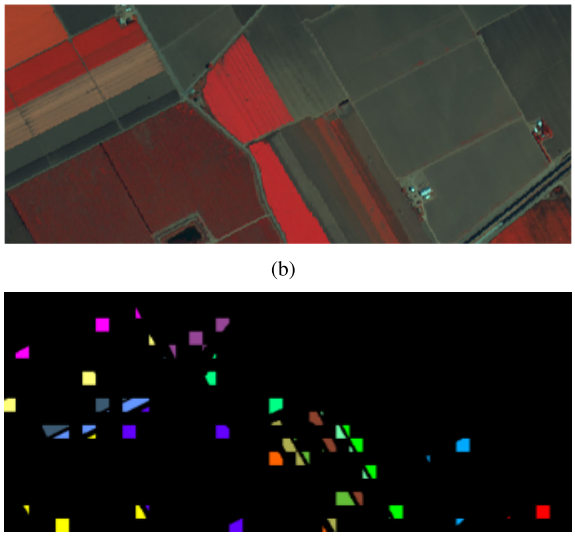


(b) (c)

Fig. 4. Indian Pines dataset. (a) Land-cover type and sample settings. (b) False-color image. (c) Spatial distribution of training samples.

ID	Color	Land Cover Type	Train	Test	Total
C1		Brocoli_green_weeds_1	59	1950	2009
C2		Brocoli_green_weeds_2	66	3660	3726
C3		Fallow	53	1923	1976
C4		Fallow_rough_plow	95	1299	1394
C5		Fallow_smooth	102	2576	2678
C6		Stubble	113	3846	3959
C7		Celery	102	3477	3579
C8		Grapes_untrained	141	11130	11271
C9		Soil_vinyard_develop	90	6113	6203
C10		Corn_senesced_green	84	3194	3278
C11		Lettuce_romaine_4wk	64	1004	1068
C12		Lettuce_romaine_5wk	73	1854	1927
C13		Lettuce_romaine_6wk	66	850	916
C14		Lettuce_romaine_7wk	71	999	1070
C15		Vinyard_untrained	85	7183	7268
C16		Vinyard_vertical_trel	44	1763	1807
		Total	2150	51979	54129

(a)



(b)

(c)

Fig. 5. Salinas dataset. (a) Land-cover type and sample settings. (b) False-color image. (c) Spatial distribution of training samples.

3) Large Dataset: The DFC 2018 dataset was acquired in 2017 by the National Center for Airborne Laser Mapping around the University of Houston campus. This dataset contains 601 × 2385 pixels with a spatial resolution of 1 m per pixel and 50 spectral bands covering a wavelength range of 380–1050 nm. On the GT, 20 types of urban land cover were defined. Fig. 6(a) summarizes the type and number of samples, while the spatial distribution of the training and test data is represented in Fig. 6(b) and (c), respectively.

The above training and test samples were all generated using the proposed CMS sampling strategy. Data augmentation techniques, such as random rotation, adding noise, and changing brightness, were applied to the training data to prevent overfitting. Before experiments, datasets were normalized to $[-1, 1]$ to harmonize the magnitude of the data.

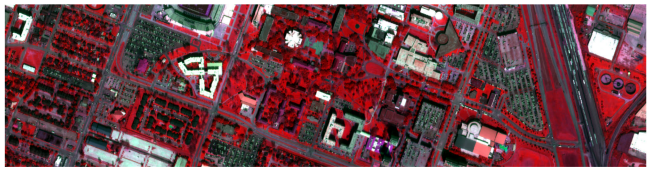
B. Experimental Setting

1) Comparison Methods: We compared the performance of our LiT with that of SVM [5], SVM with edge-preserving filtering (SVM-EPF) [62], SSRN [21], SSSAN [14], SMRN [24], SSFCN [22], ENL-FCN [54], SpectralFormer [19], ConvNeXt [63], and MobileViT [36]. SVM is a traditional pixelwise classifier, SVM-EPF is an edge-preserving filtering-based SVM classification method, and the rest are DL-based spectral–spatial feature learning algorithms. SSRN, SSSAN, and SMRN follow the patch-based classification framework. Specifically, SSRN uses successive spectral and spatial residual blocks to learn spectral and spatial representations. SSSAN employs two branches, the spatial self-attention module-based branch and the spectral self-attention module-based branch, to learn spatial and spectral information in parallel. SMRN models the latent manifold structure in each patch adaptively using the GCN. SSFCN and ENL-FCN are FCN-based architectures designed to take an entire image as input and perform pixel-to-pixel classification. SpectralFormer is a pure transformer network specifically designed for HSI classification, which takes 8 × 8 patches as input and outputs a category for the center pixel of each patch. ConvNeXt and MobileViT are SOTA backbones for vision tasks. ConvNeXt is a pure CNN model developed by ResNet according to ViT, while

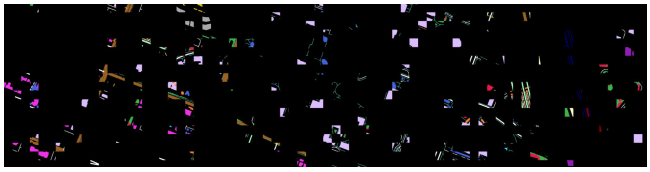
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ID	Color	Land Cover Type	Train	Test	Total
C1		Healthy grass	1201	8294	9495
C2		Stressed grass	2232	29698	31930
C3		Artificial turf	74	570	644
C4		Evergreen trees	955	12455	13410
C5		Deciduous trees	432	4496	4928
C6		Bare earth	207	4276	4483
C7		Water	145	93	238
C8		Residential buildings	2902	36603	39505
C9		Non-residential buildings	7616	216380	223996
C10		Roads	4952	39992	44944
C11		Sidewalks	3829	28304	32133
C12		Crosswalks	309	1208	1517
C13		Major thoroughfares	3942	42841	46783
C14		Highways	757	9125	9882
C15		Railways	640	6287	6927
C16		Paved parking lots	1188	10249	11437
C17		Unpaved parking lots	64	75	139
C18		Cars	781	5824	6605
C19		Trains	561	4668	5229
C20		Stadium seats	385	6405	6790
		Total	33172	470743	501015

(a)



(b)



(c)

Fig. 6. DFC 2018 dataset. (a) Land-cover type and sample settings. (b) False-color image. (c) Spatial distribution of training samples.

MobileViT is LiT for mobile devices. The parameters of these comparison methods were set according to their respective literature.

2) Implementation Details: All comparison methods were run on the PyTorch platform and trained and tested on the same sample sets. All experiments were performed on an NVIDIA GeForce RTX 3060 GPU. We conducted ten independent tests for each experiment to avoid biased estimates. Average evaluation metrics are reported to provide a comprehensive and accurate representation of our findings.

3) Metric: The overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and producer accuracy (PA) of each category were adopted to evaluate the performance of different methods.

4) Parameter Setting: We trained our LiT for 100 epochs using the AdamW optimizer [64] with a decay rate of $1e-2$, a momentum of 0.9, and a learning rate of $5e-5$. There is a 20-epoch linear warm-up followed by a cosine decay schedule. We set the input data size to 32×32 and 64×64 on the Indian Pines and Salinas, respectively, depending on the dataset size and GPU memory. The detailed architecture of LiT is summarized in Table I, where Indian Pines and Salinas datasets use the Channel-1 setting, while the DFC 2018 dataset adopts the Channel-2 setting.

TABLE I
DETAILS OF THE LIT ARCHITECTURE

Stage	Operator	Strid	Layers	Channels	
				#1	#2
-	Conv3×3	1	1	128	48
1	Fused-MBConv, K3×3	1	2	96	36
2	Fused-MBConv, K3×3	1	2	64	24
3	Transormer	-	2	64	24
4	Transormer	-	1	32	24
-	Conv1×1	1	1	16	20

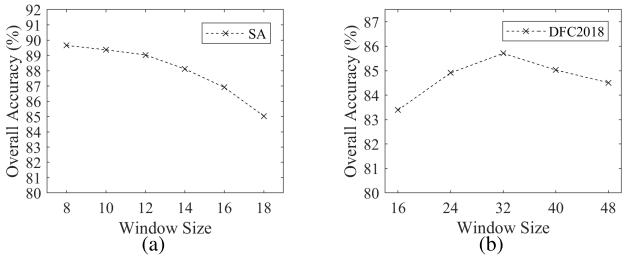


Fig. 7. Effect of window size on OA. (a) Salinas dataset. (b) DFC 2018 dataset.

We also investigated the effect of window size on the performance of our LiT. To ensure that each class is available in at least two windows, we set the window size for the Indian Pines dataset to its maximum value of 4, since there is the limited number of labeled samples for certain classes (e.g., the oats class has only 20 labeled pixels). We conducted experiments for the Salinas and DFC 2018 datasets to select the optimal window size. We investigated the generated windows fed directly into the network for training and testing on the large DFC 2018 dataset. In this way, the feature extraction spaces of the training and test samples are independent of each other, providing a more objective performance evaluation of the methods.

Fig. 7(a) shows that the accuracy decreases, as the window size decreases in the Salinas dataset. The reason for this is that a larger window size reduces the overlap between the feature extraction spaces of training and test samples and reduces the diversity of training samples. We set the window size of the Salinas to 12×12 to balance accuracy with objective benchmark.

The effect of window size on accuracy for the DFC 2018 dataset can be seen in Fig. 7(b). A large window is efficient for capturing long-range spatial information, as accuracy increases with window size. However, a too large window size decreases accuracy, since large windows reduce the number of windows and the diversity of training samples. According to Fig. 7(b), we set the window size to 32×32 for the DFC 2018 dataset.

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TABLE II
COMPARING THE PERFORMANCE OF LIT TRAINED WITH SAMPLES GENERATED BY CRS AND CMS SAMPLING STRATEGIES

Dataset	Evaluation index	CRS	CMS	Improvement
Indian Pines	Training samples	2026	2014	-12
	OA (%)	81.79	86.31	4.52
Salinas	Training samples	2176	2150	-26
	OA (%)	84.57	89.03	4.46
DFC2018	Training samples	33219	33172	-47
	OA (%)	83.15	85.41	2.26

TABLE III
ABLATION ANALYSIS OF THE PROPOSED LIT ON THE SALINAS DATASET

Method	FusedMBConv	PLSAM	CLSAM	OA (%)
(a) ViT	-	-	-	82.51
(b) CoViT	✓	-	-	85.12
(c) CoViT+PLSAM	✓	✓	✗	87.62
(d) CoViT+CLSAM	✓	✗	✓	86.76
(e) LiT	✓	✓	✓	89.03

TABLE IV
SENSITIVITY ANALYSIS OF THE MODEL LAYOUT IN TERMS OF OA (%) ON INDIAN PINES, SALINAS, AND DFC 2018 DATASETS

Model Layout	Indian Pines	Salinas	DFC2018
C-T-T-T	83.16	86.58	82.87
C-C-T-T	86.31	89.03	85.71
C-C-C-T	86.38	89.13	85.18
C-C-C-C	85.69	87.96	83.89

TABLE V
COMPARISON OF TRAINABLE PARAMETERS (PARAMS), FLOPS, TRAINING (TRAIN), AND INFERENCE (INFER) TIME, AND OA BETWEEN PATCH-BASED AND FCN-BASED FRAMEWORKS ON THE SALINAS DATASET

Framework	Params	FLOPs	Train (s)	Infer (s)	OA (%)
Patch-based	379.51K	2151.62M	2897.96	22.84	86.40
FCN-based	399.07K	6526.34M	14766.13	1.93	89.03

C. Sampling Strategy Analysis

To investigate the impact of the sampling strategy on the classification performance, we compared the performance of our LiT under the CMS sampler with that under the CRS sampler using almost the same number of training samples.

As reported in Table II, experimental results under the CMS sampler outperformed those under the CRS sampler by 4.52% on Indian Pines, 4.46% on Salinas, and 2.26% on DFC 2018, even with fewer training samples. This demonstrates that using multiclass input data can effectively enhance the learning capability of networks when compared with using single-class input data. This is due to the complementarity and collaboration between multiclass data, which facilitates a comprehensive learning process for the network. As a result, networks can handle more complex and diverse data.

D. Model Analysis

1) Ablation Studies: We conducted ablation studies on the Salinas dataset to validate the effectiveness of each component of our LiT by following a trajectory from ViT to LiT. The corresponding results are summarized in Table III.

First, we study the importance of the hybrid approach (CoViT), where the first two stages deploy the Fused-MBConv block, and the last two stages use the transformer block in ViT. As shown in Table III(b), CoViT improved OA from 82.51% to 85.12% compared with ViT. Fused-MBConv blocks can effectively learn representations of local features in early layers due to their strong inductive biases. This suggests that deploying convolutions in ViT for early vision processing may be a critical design decision that makes a trade-off between inductive biases and the representational learning capability of transformer blocks.

Second, to evaluate the proposed PLMSA and CLMSA modules, we conducted experiments by replacing the MSA in CoViT with them individually. Table III(c) shows that the PLMSA improved OA from 85.12% to 87.62%. CLMSA alone outperformed CoViT by 1.64%, as illustrated in Table III(d). Joint exploitation of PLMSA and CLMSA resulted in further performance improvements, as evident in Table III(e). These results indicate that constructing relationships between individual pixels/channels and eigenvectors in the spectral/spatial dimension is valuable for exploiting the global context over local features.

2) Model Layout Analysis: The performance of models is heavily influenced by its layout. Therefore, it is necessary to investigate the most effective configuration of LiT. Given that convolution is better suited for processing local features in early stages, we imposed the restriction that convolution stages must precede transformer stages. With this constraint, we investigated either the Fused-MBConv or transformer block from stages 2 to 4. This results in four versions: C–T–T–T, C–C–T–T, C–C–C–T, and C–C–C–C, where C and T denote convolution and transformer blocks, respectively. We compared the performance of these four layouts.

Table IV shows that C–T–T–T produced the lowest OA (83.16% on Indian Pines, 86.58% on Salinas, and 82.87% on DFC 2018), followed by C–C–C–C (85.69% on Indian Pines, 87.96% on Salinas, and 83.89% on DFC 2018), whereas C–C–T–T and C–C–C–T produced comparable results. C–C–T–T achieved the best results on the DFC 2018 dataset, while C–C–C–T produced the best results on Salinas and Indian Pines datasets, yet the difference between them is minor. Considering that the transformer has strong adaptability to extensive data, we adopt C–C–T–T for LiT to make it more suitable for large-scale datasets. Moreover, the fact that C–C–T–T is better than C–T–T–T suggests that convolution may be more efficient at processing low-level information than self-attention mechanisms while using significantly less computation and memory.

3) Analysis of Learning Frameworks: This section investigates the impact of different learning frameworks (i.e., patch-based classification and FCN-based segmentation) on performance. To ensure a fair comparison of model complexity and speed, we directly replaced the last convolutional layer in LiT with a global average pooling followed by a linear layer to represent the patch-based

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TABLE VI
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Metrics	SVM [5]	SVM-EPF [62]	SSRN [21]	SSSAN [14]	SMRN [24]	SSFCN [22]	ENL-FCN [22]	SpectralFormer [19]	ConvNeXt [63]	MobileViT [36]	LiT (Ours)
OA (%)	76.67 (0)	82.94 (0)	80.77 (2.56)	84.16 (2.31)	85.71 (2.21)	79.93 (2.09)	83.02 (1.98)	77.33 (3.06)	85.19 (2.29)	83.62 (2.68)	**86.31** (1.88)
AA (%)	76.92 (0)	81.78 (0)	73.57 (3.26)	76.41 (2.70)	76.43 (2.70)	78.69 (3.14)	81.61 (3.70)	72.86 (1.96)	82.59 (2.69)	79.55 (2.56)	**79.32** (1.49)
$\kappa \times 100$	73.29 (0)	79.96 (0)	77.62 (2.81)	82.41 (2.52)	83.51 (2.33)	76.73 (2.29)	80.45 (2.16)	73.50 (1.56)	82.87 (2.86)	81.13 (2.06)	**84.13** (2.20)
C1	72.31	78.65	36.46	11.46	23.96	68.75	51.56	25.00	78.65	56.25	**80.59**
C2	74.18	80.15	78.76	**88.35**	88.16	68.92	71.06	65.37	82.64	78.91	79.08
C3	63.45	74.64	70.21	82.77	76.65	**96.14**	65.94	67.39	66.64	78.02	66.67
C4	43.27	49.31	67.97	78.39	89.58	92.19	80.27	48.44	71.04	81.25	**95.90**
C5	82.45	94.08	95.59	92.33	94.89	**97.56**	93.55	93.73	87.56	89.55	95.82
C6	89.85	95.11	**100**	99.19	**100**	70.66	99.63	97.31	93.29	98.53	99.27
C7	**94.78**	98.56	79.17	66.67	47.92	62.50	75.00	81.25	**100**	75.23	60.94
C8	91.85	92.24	94.19	94.19	95.05	72.26	93.23	99.03	90.62	99.68	**100**
C9	87.12	88.65	15.68	30.67	33.33	33.33	**79.17**	66.67	55.56	16.67	54.17
C10	68.95	72.46	56.97	78.43	74.29	66.34	**84.07**	51.31	75.83	83.17	81.33
C11	75.03	82.72	81.91	**89.56**	82.95	91.15	79.76	81.81	91.26	76.01	86.25
C12	67.69	70.81	54.94	**98.95**	91.75	92.33	84.15	59.58	72.32	74.56	87.37
C13	94.79	96.68	99.71	97.94	**100**	**100**	**100**	92.92	97.91	**100**	**100**
C14	96.72	98.42	97.77	**99.12**	88.77	63.783	95.04	98.27	95.06	96.89	96.74
C15	50.54	57.08	62.79	68.61	77.78	**87.30**	75.26	46.03	75.14	68.25	67.46
C16	77.80	78.89	84.72	44.44	75.00	95.83	78.13	91.67	87.96	**100**	89.58

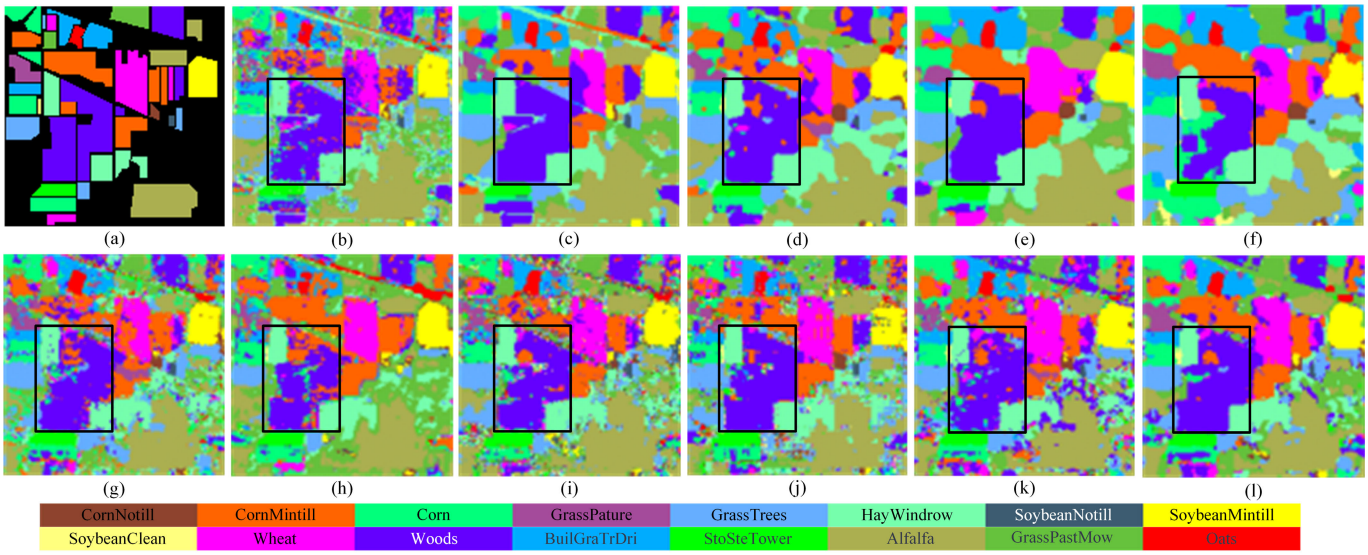


Fig. 8. Classification maps of different methods on the Indian Pines dataset. (a) GT. (b) SVM. (c) SVM-EPF. (d) SSRN. (e) SSSAN. (f) SMRN. (g) SSFCN. (h) ENL-FCN. (i) SpectralFormer. (j) ConvNeXt. (k) MobileViT. (l) LiT.

classification. Both learning frameworks were trained on the same training set and tested on the same test set. The number of trainable parameters (Parmas) and FLOPs was used to measure model complexity and theoretical computational overhead, respectively. Meanwhile, training time and inference time were used to represent actual efficiency. The corresponding results on the Salinas dataset are summarized in Table V. Table V shows that the number of parameters in both frameworks is almost equal. The FCN-based framework is less efficient than the patch-based framework in terms of FLOPs and training time. However, the FCN-based framework achieved higher accuracy (89.03% versus 86.40%) and faster inference speed (1.93 versus 22.84 s) than the patch-based classification. Although the FCN-based framework consumed more training time than the patch-based framework, the actual computation of the FCN-based framework is faster than that of the patch-based framework. This is because the time-consuming training process is performed offline, while inference speed is the main factor determining whether a method is practical. Therefore, the patch-based framework is a better choice when the amount of data is small, while the FCN-based framework is more suitable for large-scale applications (e.g., classification on satellite images) or real-time applications (e.g., those on mobile devices).

E. Comparison With Other Methods

1) Quantitative Evaluation: Tables VI–VIII summarize the mean and standard deviation of OA, AA, and κ, as well as

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TABLE VII
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON THE SALINAS DATASET

Metrics	SVM [5]	SVM-EPF [62]	SSRN [21]	SSSAN [14]	SMRN [24]	SSFCN [22]	ENL-FCN [22]	SpectralFormer [19]	ConvNeXt [63]	MobileViT [36]	LiT (Ours)
OA (%)	79.54 (0)	85.34 (0)	86.48 (2.28)	86.65 (1.94)	85.74 (1.93)	87.58 (1.86)	85.34 (1.46)	83.56 (1.62)	88.02 (1.76)	85.40 (2.19)	**89.03** (1.29)
AA (%)	86.44 (0)	89.32 (0)	94.93 (2.57)	95.45 (2.57)	93.75 (2.18)	92.93 (2.48)	89.32 (2.01)	89.68 (2.70)	**94.00** (2.46)	86.97 (2.39)	93.52 (0.96)
$\kappa \times 100$	76.99 (0)	83.47 (0)	84.97 (2.53)	85.22 (2.07	84.14 (2.24)	86.07 (2.02)	83.47 (1.53)	81.53 (1.72)	86.66 (1.81)	83.55 (2.52)	**87.73** (1.56)
C1	**100**	96.00	**100**	**100**	94.26	86.31	96	93.16	98.69	**100**	**100**
C2	75.12	99.23	99.80	**100**	99.80	99.72	99.23	87.42	99.80	99.89	99.66
C3	74.16	58.63	95.77	**100**	99.19	88.00	58.63	81.08	74.70	50.43	63.95
C4	99.36	99.45	99.64	99.54	**99.73**	97.72	99.45	98.54	99.38	80.20	99.54
C5	44.91	97.62	99.57	91.66	98.51	98.98	97.62	71.95	99.51	76.93	**99.66**
C6	**100**	99.81	**100**	99.89	99.61	99.94	99.81	99.97	**100**	**100**	**100**
C7	**100**	99.38	**100**	**100**	99.94	99.88	99.38	99.94	99.94	99.47	**100**
C8	88.48	93.02	52.70	42.35	51.80	82.77	**93.02**	85.62	61.80	92.92	78.65
C9	99.97	99.41	**100**	**100**	99.24	**100**	99.41	99.98	99.90	**100**	99.98
C10	94.20	91.26	**96.90**	96.27	91.29	94.06	91.26	94.13	94.53	92.68	94.29
C11	99.07	85.93	98.95	**99.54**	95.93	96.98	85.93	98.02	98.69	94.07	99.19
C12	**100**	99.58	**100**	99.94	99.82	**100**	99.58	**100**	99.97	**100**	**100**
C13	99.45	97.81	**100**	99.86	99.86	**100**	97.81	99.73	99.18	95.21	**100**
C14	99.16	90.96	97.59	**100**	92.29	95.42	90.96	97.29	95.72	72.29	96.51
C15	12.01	27.89	79.88	**98.37**	81.48	48.98	27.89	30.50	84.29	37.91	67.79
C16	97.14	93.15	98.15	**99.70**	97.32	98.09	93.15	97.62	97.84	99.46	97.02

TABLE VIII
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON THE DFC 2018 DATASET

Metrics	SVM [5]	SVM-EPF [62]	SSRN [21]	SSSAN [14]	SMRN [24]	SSFCN [22]	ENL-FCN [22]	SpectralFormer [19]	ConvNeXt [63]	MobileViT [36]	LiT (Ours)
OA (%)	68.87 (0)	76.04 (0)	81.89 (2.15)	82.23 (1.46)	82.74 (1.42)	82.08 (1.81)	80.91 (1.61)	77.29 (1.69)	82.81 (1.69)	82.35 (1.98)	**85.71** (1.19)
AA (%)	53.96 (0)	66.78 (0)	78.42 (2.40)	82.99 (1.70)	75.93 (2.43)	77.36 (2.19)	79.31 (2.23)	72.07 (1.49)	74.18 (2.64)	73.93 (2.96)	**83.581** (1.23)
$\kappa \times 100$	58.97 (0)	68.07 (0)	76.47 (2.33)	77.05 (1.53)	77.17 (1.60)	76.77 (1.91)	74.81 (1.74)	70.75 (2.17)	77.28 (1.52)	76.05 (1.81)	**81.04** (1.37)
C1	83.44	85.68	75.71	88.29	87.73	83.44	**93.80**	83.52	79.21	71.71	87.03
C2	93.71	94.14	93.29	90.17	93.10	**94.01**	85.06	85.51	92.72	74.01	88.01
C3	**100**	**100**	**100**	**100**	99.81	97.28	99.81	98.50	52.53	98.05	98.63
C4	94.23	**96.35**	93.88	91.02	94.79	83.97	95.90	95.61	93.01	77.19	92.39
C5	3.23	34.49	79.27	**82.52**	67.62	73.83	54.65	48.37	64.42	69.01	76.53
C6	0.32	29.52	89.58	68.79	79.62	83.65	86.86	61.86	85.12	74.18	**90.22**
C7	73.68	78.95	69.47	**100**	**100**	**100**	**100**	86.84	93.68	68.71	97.85
C8	76.16	84.40	79.90	92.28	88.15	86.13	71.41	87.09	89.74	**93.06**	92.58
C9	77.77	86.93	83.25	80.58	87.79	85.17	87.35	79.87	87.11	**92.98**	89.32
C10	54.27	58.51	75.61	78.00	73.92	79.22	72.95	60.73	**80.17**	74.68	77.59
C11	51.14	72.55	**76.37**	74.79	51.55	68.18	59.16	66.36	73.82	57.91	66.63
C12	0	4.92	15.27	5.84	4.30	15.03	3.97	14.65	16.91	**23.63**	12.49
C13	43.98	61.80	70.00	**81.92**	67.71	62.33	64.38	67.12	64.90	72.33	73.72
C14	29.47	60.81	68.16	67.19	65.55	64.65	78.01	65.26	59.23	47.59	**81.90**
C15	82.66	83.32	98.20	95.84	94.19	**98.95**	92.43	95.94	96.36	95.60	95.53
C16	29.45	54.25	91.23	79.58	87.33	94.89	88.66	76.82	**96.52**	91.36	91.50
C17	0	31.43	28.57	**100**	0.95	10.26	89.66	9.86	6.59	73.69	88.00
C18	49.12	68.81	93.70	**98.00**	92.96	87.70	78.41	81.79	69.78	79.64	87.72
C19	44.66	68.87	91.89	**97.68**	84.54	87.45	84.83	83.13	87.84	86.86	87.35
C20	91.99	93.42	94.97	87.34	97.02	91.37	**98.85**	92.36	93.96	56.50	96.65

the average PA for each class. The best performance in each row is highlighted in bold.

As a conventional classifier, SVM exhibits the worst results, especially on the DFC 2018 dataset. SVM considers raw pixel vectors as features and feeds them directly into classifiers, which cannot excavate the discriminative information contained in the data. SVM-EPF dramatically improves the performance of SVM by applying edge-preserving filtering. With the introduction of spatial information and the strong learning capability of DL techniques, patch-based networks, such as SSRN, SSSAN, and SMRN, also significantly outperform SVM. Although these patch-based methods have achieved remarkable accuracy, their limited patch size limits the spatial context and tends to lose some important information. FCN-based networks (i.e., SSFCN and ENL-FCN) only achieved on-par results with patch-based networks (i.e., SSRN,

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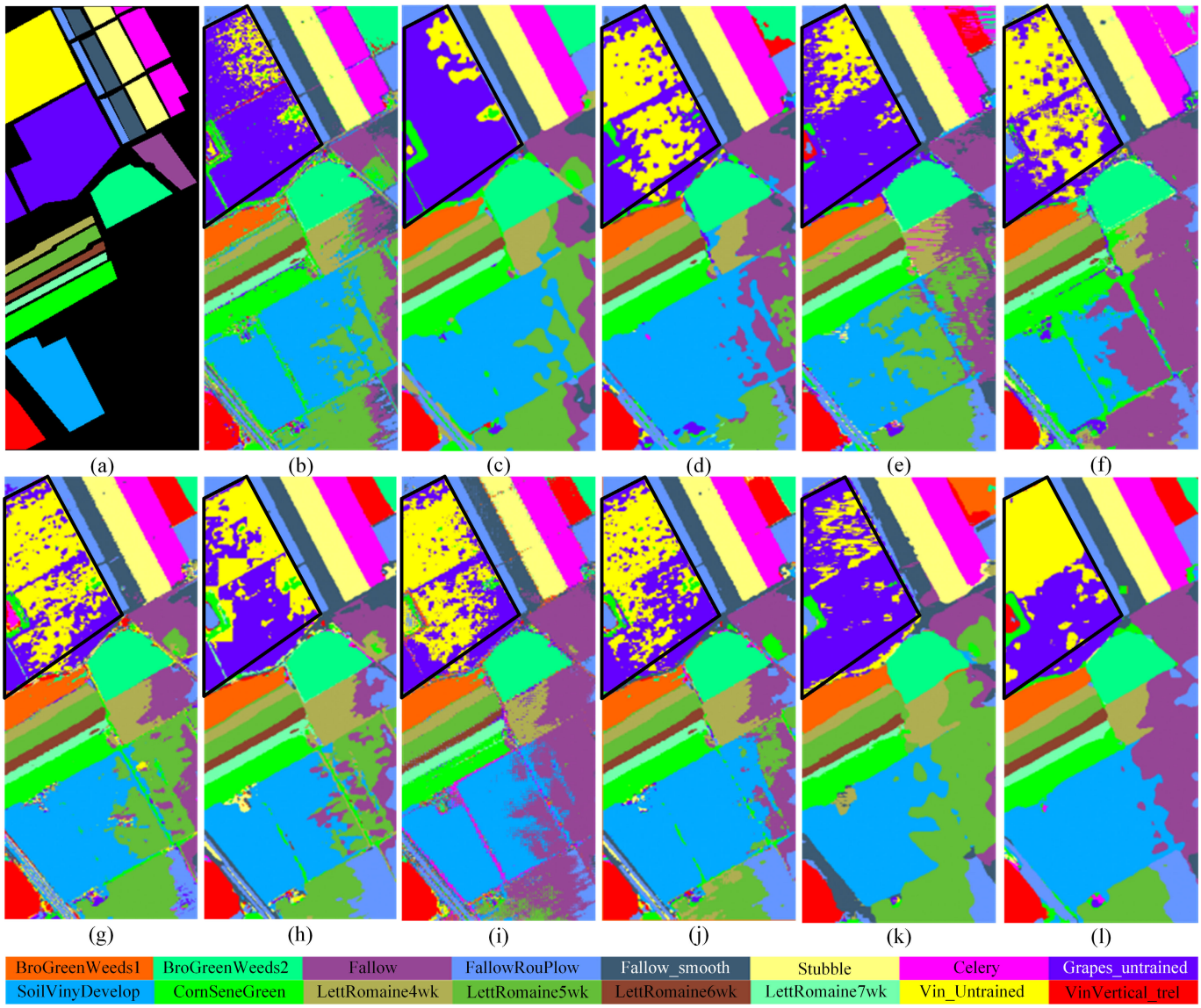


Fig. 9. Classification maps of different methods on the Salinas dataset. (a) GT. (b) SVM. (c) SVM-EPF. (d) SSRN. (e) SSSAN. (f) SMRN. (g) SSFCN. (h) ENL-FCN. (i) SpectralFormer. (j) ConvNeXt. (k) MobileViT. (l) LiT.

SSSAN, and SMRN) under the proposed sampling strategy. For example, on the Salinas dataset, SSFCN and ENL-FCN achieved 87.58% and 85.34%, respectively, while SSRN, SSSAN, and SMRN achieved 86.48%, 86.65%, and 85.74%, respectively. These FCN-based networks mainly use a stack of many small spatial convolutions (e.g., 3×3) to enlarge the receptive fields. According to the effective receptive field (ERF) theory, the size of the ERF is proportional to $O(K\sqrt{L})$, where K is the kernel size and L is the depth, i.e., the number of layers [65]. Thus, the ERF of FCN-based networks is still limited, which may limit their performance gains. Although the transformer is good at modeling global dependencies, SpectralFormer is inferior to CNN-based or GCN-based models, since it lacks inductive biases built-in CNNs and still follows the patch-based classification. As the SOTA backbones, ConvNeXt performs well on all three datasets, while MobileViT's results are not satisfactory. Although MobileViT employs both local information and global connectivity to reason about the relationships between image contents, it is still heavy weight for HSI datasets.

The proposed LiT exhibited a significant improvement over all the comparison algorithms, because it uses transformers to capture long-range dependencies and convolution to gather local information. Both long-range connectivity and local information contribute to reasoning about the relationships between image contexts. For example, LiT achieved a maximum OA of 89.03% on the Salinas dataset, outperforming SVM, SSRN, SSSAN, SMRN, SSFCN, ENL-FCN, SpectralFormer, ConvNeXt, and MobileViT by 9.49%, 2.55%, 2.38%, 3.29%, 1.45%, 3.69%, 5.47%, 1.01%, and 3.63%, respectively. Even for some challenging classes with similar spectral and textural features (e.g., stubble and grapes_untrained on the Salinas dataset), the proposed LiT still achieved better accuracy by using its dynamic TwinMSA modules to capture complicated relational interactions between different positions. These results powerfully demonstrate the practicality and value of our LiT.

2) Visual Evaluation: Figs. 8–10 show the classification maps produced by the comparison methods and the corresponding GT images. We can see that these qualitative

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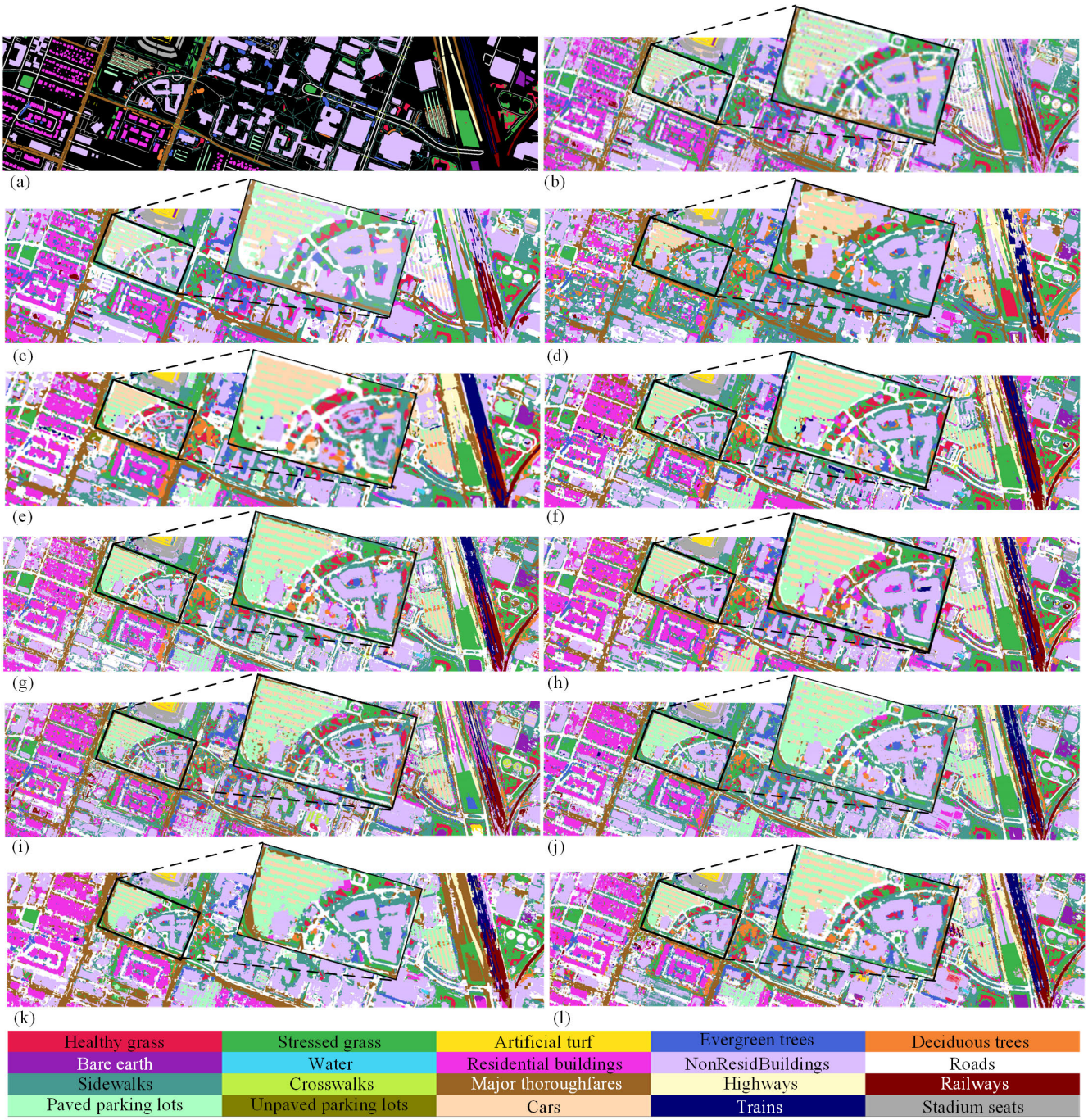


Fig. 10. Classification maps of different methods on the DFC 2018 dataset. (a) GT. (b) SVM. (c) SVM-EPF. (d) SSRN. (e) SSSAN. (f) SMRN. (g) SSFCN. (h) ENL-FCN. (i) SpectralFormer. (j) ConvNeXt. (k) MobileViT. (l) LiT.

comparisons between these methods are in agreement with the quantitative comparisons presented in Tables VI–VIII.

As shown in Figs. 8(c), 9(c), and 10(c), the classification maps of SVM exhibit a considerable salt–pepper noise, because SVM only considers spectral information. Due to the similar spectral information between certain classes (e.g., grapes_untrained and vinyard_untrained on the Salinas dataset), distinguishing these easily confused categories using spectral information alone is difficult. SVM-EPF not only makes object edges sharper, but also significantly mitigates the pepper noise problem of SVM, especially on the Indian Pines dataset. The patch-based networks, such as SSRN, SSSAN, and SMRN, also produced better visual performance than SVM by incorporating the spatial features of neighboring pixels. Classification maps generated by patch-based networks contain fewer noisy points and are smoother. However, the boundaries of these classification maps tend to be distorted, since patch-based classification approaches assume that each pixel within a patch contributes equally to classifying the center pixel. This assumption is valid for most pixels in

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TABLE IX
COMPARISON OF PARAMS, FLOPS, TRAINING (TRAIN), AND INFERENCE (INFER) TIME OF DIFFERENT METHODS ON THE SALINAS DATASET

	Method	Params	FLOPs	Train(s)	Infer(s)
Patch	SSRN	370.31K	7690.14M	475.85	31.09
	SSSAN	389.62K	2579.04M	571.58	48.65
	SMRN	808.51K	1292.67M	775.82	23.74
	SpectralFormer	351.00K	2300.19M	3897.56	26.59
FCN	SSFCN	110.26K	3612.87M	5206.23	0.59
	ENL-FCN	139.75K	4579.16M	16351.33	2.66
	ConvNeXt	534.12K	7122.60M	17985.63	2.96
	MobileViT	981.68K	8110.45M	18689.26	3.06
	LiT (Ours)	399.07K	6526.34M	14766.13	1.93

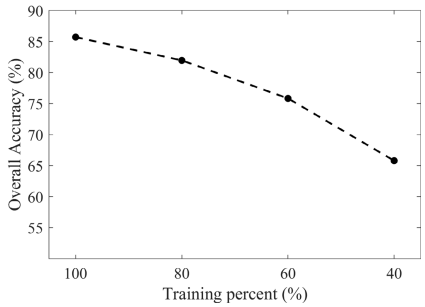


Fig. 11. Classification accuracy of LiT with different numbers of training samples on the DFC 2018 dataset.

homogeneous regions, while spatial support from adjacent pixels is often ineffective or even disruptive when belonging to different categories. Compared with patch-based algorithms, SSFCN, ENL-FCN, and ConvNeXt obtained slightly better boundary localization. As seen, classification maps generated by CNNs tend to smooth or overlook details, such as class boundaries. This results from the local connectivity and parameter-sharing properties of CNNs, which limit their ability to detect fine details and accurately localize objects. There are more misclassifications in the classification maps of SpectralFormer and MobileViT than in ConvNeXt. Our LiT produced high-quality classification maps with well-positioned boundaries and smooth objects, while the object outline of LiT is still inferior to SVM and SVM-EPF. This is mainly because SVM did not consider spatial information, while LiT did. SVM-EPF produced more accurate edges than SVM by optimizing the classification maps of SVM-EPF.

The experimental results also show that transformers can be as effective as CNNs, even with small-scale datasets. This can be attributed to the inductive bias inherent in convolutional layers, which results in higher generalization and faster convergence speed. On the other hand, transformer layers have a larger model capacity that can benefit from larger datasets. Therefore, combining convolutional and attention layers can achieve better generalization and capacity.

3) Model Complexity and Speed Evaluation: To comprehensively analyze the efficiency of our LiT, we compared it with other methods in terms of Params and FLOPs, as well as training and inference time on the SA dataset. As listed in Table IX, LiT obtained on par Params and FLOPs with the average level of other methods, while having a faster inference speed. In general, the FCN-based methods take longer training time but less inference time than the patch-based methods. This again demonstrates that the FCN-based framework is more suitable for real-world applications. Among these FCN-based models, LiT has more Params and FLOPs than SSFCN and ENL-FCN but less than ConvNeXt and MobileViT. In addition, the inference speed of our LiT is much faster and achieves near real-time inference, demonstrating its potential for efficient hyperspectral scene analysis.

4) Extended Experiment: To further evaluate the performance of the proposed LiT, we first compared it with other methods on the Salinas dataset using samples generated by the random sampling strategy. The numbers of training and test samples were kept the same as in Fig. 5(a). We also used OA, AA, and κ for performance evaluation, and the corresponding results are presented in Table X. The table shows that the proposed method also yielded the best results. SSSAN, SMRN, SSFCN, ENL-FCN, SpectralFormer, ConvNeXt, MobileViT, and LiT all achieved high OA (even close to 100%) with little difference between each other. The reason is that there is a large overlap in the feature extraction spaces between the training and test samples. Information from training data was used to evaluate the methods during the testing phase, resulting in exaggerated results. Therefore, it is imperative to establish a standardized performance evaluation method to ensure consistent and reliable results. The proposed CMS sampling strategy provides a solution doing this direction.

Then, we compared our LiT with SVM with region-based max-voting (SVM-RMV) [66], FasterNet [67], and EfficientViT [48] on the Salinas dataset using samples in Fig. 5. SVM-RMV, an edge-preserving method, applies the region-based max-voting scheme on the SVM classification map to optimize classification maps. FasterNet is a neural network built upon partial convolution, and EfficientViT is a ViT, both of which achieve SOTA speed and accuracy trade-offs on various vision tasks. Results in Table XI show that our LiT still delivers the best results. FasterNet and EfficientViT are superior to SVM-RMV, confirming that FasterNet and EfficientViT have better generalization and performance than SVM-RMV. LiT shows better results than FasterNet and EfficientViT, again demonstrating its superiority and generalizability again.

F. Impact of the Number of Training Samples

The performance of DL-based supervised methods is strongly influenced by the number of training samples. To investigate the sensitivity of LiT to the number of training samples, we randomly selected different numbers of training samples from the training set of the DFC 2018 dataset (as shown in Fig. 6) in the proportion of [100%, 80%, 60%, 40%]. The corresponding results are summarized in Fig. 11. We can see that the classification accuracy gradually decreases, as the percentage of training samples decreases. However, the decline is relatively slow, demonstrating the stability and robustness of the proposed LiT.

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TABLE X
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON THE SALINAS DATASET USING A RANDOM SAMPLING STRATEGY

Metrics	Methods										
	SVM [5]	SVM-EPF [62]	SSRN [21]	SSSAN [14]	SMRN [24]	SSFCN [22]	ENL-FCN [22]	SpectralFormer [19]	ConvNeXt [63]	MobileViT [36]	LiT (Ours)
OA (%)	81.76	87.65	97.65	98.79	98.66	98.16	98.42	97.24	98.24	98.01	99.01
AA (%)	87.79	89.68	98.96	99.12	99.56	98.96	99.16	97.57	98.01	97.89	99.23
$\kappa \times 100$	79.29	85.67	97.28	98.87	98.65	97.98	98.27	97.13	98.19	98.05	98.97

TABLE XI
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON THE SALINAS DATASET USING OUR CMS SAMPLING STRATEGY

Methods	OA (%)	AA(%)	$\kappa \times 100$
SVM-RMV [66]	86.26	88.69	85.69
FasterNet [67]	88.56	92.61	87.01
EfficientViT [48]	87.89	92.78	86.94
LiT (Ours)	89.03	93.52	87.73

V. CONCLUSION

This article has proposed a hybrid architecture for HSI classification, called LiT, that can effectively address the limitations of using transformers on small-scale datasets. LiT improves the representational ability of the extracted features and achieves a favorable accuracy–speed trade-off by combining the strengths of transformers and CNNs. Moreover, the proposed PLMSA and CLMSA modules significantly enhanced long-range contextual information and successfully extracted more discriminative features in complex regions, while consuming less computational and memory resources. The proposed CMS sampling strategy significantly reduced the overlap of feature extraction regions between training and test samples. Extensive experiments demonstrated that the LiT outperforms other SOTA classification methods. Moreover, the experiments also revealed that selecting training samples from multiclass regions can make the trained network more discriminative, and the FCN-based segmentation framework is better for real-world, large-scale applications compared with the patch-based classification framework. In the future, we will focus on constructing large benchmark datasets to facilitate spectral–spatial HSI analysis. In addition, weakly supervised and self-supervised approaches should be considered to reduce the need for expensive pixel-level image annotation.

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